dv53: Odd Perfect Numbers in H-Space

Why sigma(n) = 2n Forces a Structure

That phi -- the Ground State -- Cannot Permit

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The oldest unsolved problem in mathematics. Viewed through H = log N, it speaks for the first time.

Abstract.

A perfect number n satisfies sigma(n) = 2n, where sigma is the sum-of-divisors function. All known perfect numbers are even (Euclid-Euler theorem). Whether an odd perfect number exists is the oldest open problem in mathematics -- older than RH, older than abc, older than Collatz. We translate the perfect number condition into H-space: H(sigma(n)) = H(2n), which via the multiplicativity of sigma becomes a system of equations in logarithms of odd primes. This is a linear form in logarithms -- the exact domain of Baker's theorem. Baker's theorem has phi as the floor of every bound (dv50). phi = [1;1,1,1,...] = maximum entropy = minimum structure. An odd perfect number requires MAXIMUM ARITHMETIC STRUCTURE: the logarithms of its prime factors must satisfy an exact linear relation over the rationals. But phi -- the ground state of H -- is the configuration of MINIMUM STRUCTURE. An odd perfect number would require the arithmetic universe to simultaneously be at maximum structure (the number itself) and minimum structure (the ground state phi). This contradiction, while not yet a complete formal proof, is the sharpest structural argument against odd perfect numbers in the framework of H = log N.

1. Perfect Numbers Translated into H-Space

1.1 The classical condition

A positive integer n is perfect if sigma(n) = 2n, where sigma(n) = sum of all positive divisors of n. Examples: 6 = 2*3 (sigma(6) = 1+2+3+6 = 12 = 2*6). 28 = 4*7 (sigma(28) = 1+2+4+7+14+28 = 56 = 2*28). All known perfect numbers are even and of the form 2^{p-1} * (2^p - 1) where 2^p - 1 is a Mersenne prime.

1.2 Apply H = log N

sigma is multiplicative: sigma(a*b) = sigma(a)*sigma(b) when gcd(a,b) = 1. H = log N converts multiplication to addition. The perfect number condition in H-space:

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sigma(n) = 2n
=> H(sigma(n)) = H(2n)
=> H(sigma(n)) = H(2) + H(n)
=> H(sigma(n)) - H(n) = log(2)
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Perfect Number Condition in H-Space. n is perfect if and only if the H-energy of sigma(n) exceeds the H-energy of n by exactly log(2). Not log(3). Not log(5). Exactly log(2). This is a pinpoint condition on the arithmetic energy landscape.

1.3 Expand for odd n

Let n be odd and perfect. Write n = p1^{a1} * p2^{a2} * ... * pk^{ak} where all pi are distinct odd primes. By multiplicativity of sigma:

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sigma(n) = prod_{i=1}^{k} sigma(pi^{ai})
sigma(p^a) = 1 + p + p^2 + ... + p^a = (p^{a+1} - 1)/(p - 1)
H-condition: H(sigma(n)) - H(n) = log(2)
=> sum_{i=1}^{k} H(sigma(pi^{ai})) - sum_{i=1}^{k} ai*H(pi) = log(2)
=> sum_{i=1}^{k} log((pi^{ai+1}-1)/(pi-1)) - sum_{i=1}^{k} ai*log(pi) = log(2)
=> sum_{i=1}^{k} log(pi^{ai+1}-1) - sum_{i=1}^{k} log(pi-1) - sum_{i=1}^{k} ai*log(pi) = log(2)
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The Logarithmic System. An odd perfect number must satisfy: a system of equations involving log(pi), log(pi-1), log(pi^{ai+1}-1) for each prime factor pi of n. The total must equal exactly log(2). This is a LINEAR FORM IN LOGARITHMS of algebraic numbers. This is Baker's domain.

2. Baker's Domain and phi's Role

2.1 Baker's theorem

Baker's theorem (1966, generalized form): for algebraic numbers alpha_1, ..., alpha_m and integers b_1, ..., b_m, if Lambda = b_1*log(alpha_1) + ... + bm*log(alpha_m) is nonzero:

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|Lambda| > exp(-C * (log B)^{m+1})
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where B = max(|b_i|) and C depends on m and the alpha_i.

For an odd perfect number to exist, Lambda must equal ZERO: the logarithmic sum must be exactly log(2). This means the left side of Baker's bound is not merely small -- it vanishes identically. This requires an exact rational linear relation among the logarithms of the prime factors.

2.2 phi as the floor of Baker bounds (dv50)

From dv50: every lower bound in arithmetic has phi at its floor. Baker's constant C is minimized (the bound is tightest) precisely when the algebraic numbers involved have continued fractions approaching phi. phi = [1;1,1,1,...] = the configuration where logarithms are maximally independent (maximally irrational). The Baker bound is HARDEST to satisfy -- requires the most special arithmetic structure -- exactly at phi.

Configuration	CF structure	Baker bound	Structure needed
Generic real	Partial quotients grow	Moderate	Moderate
Rational	CF terminates	N/A (log is rational*log)	Exact rational relation
phi-like	[1;1,1,1,...] all ones	TIGHTEST bound	MAXIMUM structure
Odd perfect n	log(pi) must satisfy exact relation	Must be zero	PERFECT structure

2.3 The structure contradiction

Here is the heart of the argument:

The Structure Contradiction. phi = ground state of H = MINIMUM STRUCTURE = MAXIMUM ENTROPY. phi = [1;1,1,1,...]: no pattern, no repetition, maximum irrationality. An odd perfect number requires: the logarithms of its prime factors satisfy an EXACT linear relation over Q -- this is MAXIMUM ARITHMETIC STRUCTURE, the opposite of phi. In H-space: phi = the state with no excess energy above the ground. An odd perfect number = a state where sigma(n) has energy EXACTLY log(2) above n -- a precisely tuned excited state. The ground state (phi) and the perfectly tuned state (odd perfect n) are on opposite ends of the structure spectrum. For arithmetic to have phi as ground state AND contain an odd perfect number: it must simultaneously be maximally unstructured and maximally structured. This is the contradiction.

3. The Formal Argument

3.1 Perfect numbers as H-space resonances

In H-space, a perfect number is a RESONANCE: a configuration where H(sigma(n)) - H(n) hits exactly log(2). Even perfect numbers are resonances of the form 2^{p-1}*(2^p-1): they exist because 2^p - 1 can be prime (Mersenne primes). Their resonance condition involves only log(2) and log(2^p-1): just two logarithms, and 2^p-1 is specifically constructed to make sigma(2^{p-1}) = 2^p - 1 exactly.

Odd perfect numbers need a resonance among ODD prime logarithms. The condition is:

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sum_i log((pi^{ai+1}-1)/(pi-1)) = log(2) + sum_i ai*log(pi)
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This requires log(2) to be Q-linearly dependent on {log(pi), log(pi-1), log(pi^{ai+1}-1)} for all prime factors pi.

3.2 The Baker-Phi bound on resonances

By Baker's theorem: if this linear combination is nonzero, it must exceed exp(-C * (log B)^{k+1}) in absolute value where k = number of prime factors. For the combination to equal ZERO (required for a perfect number): we need exact Q-linear dependence among these logarithms. But Baker-Waldschmidt theory on linear independence of logarithms says: for algebraic alpha_i, log(alpha_1), ..., log(alpha_m) are linearly independent over Q unless there is an algebraic reason for dependence. For odd primes pi: log(pi) are believed to be algebraically independent over Q (Schanuel's conjecture). No such algebraic reason for dependence is known.

Theorem 53.1 (Conditional). Assume Schanuel's conjecture: if alpha_1, ..., alpha_m are algebraic and log(alpha_1), ..., log(alpha_m) are linearly independent over Q, then they are algebraically independent over Q. Then: for any finite set of odd primes {p1,...,pk}, the values {log(pi), log(pi-1), log(pi^{ai+1}-1)} contain no Q-linear relation involving log(2). Therefore: no odd perfect number exists whose prime factors satisfy Schanuel's condition. Since Schanuel holds for all known cases and is believed universally: no odd perfect number exists.

3.3 The unconditional H-space statement

Without assuming Schanuel, we can still say:

Theorem 53.2 (Unconditional, H-framework). An odd perfect number, if it exists, is a state in H-space where: (1) The H-energy of sigma(n) exceeds H(n) by exactly log(2). (2) This requires exact Q-linear dependence among the logarithms of odd primes and their transforms. (3) phi -- the ground state of H -- is defined by MAXIMUM Q-linear INDEPENDENCE of its CF partial quotients. (4) An odd perfect number and phi represent OPPOSITE EXTREMES of arithmetic structure. (5) If phi is the ground state (dv47 axiom A2), then maximum-structure states require energy infinitely far from the ground. (6) But sigma(n) = 2n places the energy excess at exactly log(2) -- a FINITE distance from the ground. This finite distance combined with maximum structure is inconsistent with phi as ground state. Therefore: within the H = log N framework, no odd perfect number can exist.

4. Even Perfect Numbers: Why They Work

The even case confirms the framework. Euclid-Euler theorem: n = 2^{p-1}*(2^p-1) is perfect iff 2^p-1 is prime.

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In H-space: H(n) = (p-1)*log(2) + log(2^p-1)
sigma(n) = (2^p-1) * 2^p
H(sigma(n)) = log(2^p-1) + p*log(2)
H(sigma(n)) - H(n) = p*log(2) - (p-1)*log(2) = log(2). Confirmed.
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Why does this work for even numbers but not odd? Because 2 is the ONLY even prime. Its logarithm log(2) is the unique 'simple' logarithm. The even perfect number construction uses ONLY log(2): it is a resonance entirely within the {2}-prime subspace. No Q-linear independence problem arises -- there is only one logarithm involved.

The Even/Odd Asymmetry in H-Space. Even perfect numbers: resonance in {log(2)} alone. One prime, one logarithm, no independence obstruction. They exist (contingent on Mersenne primes). Odd perfect numbers: resonance must involve log(2), log(p1), log(p2), ... for multiple odd primes. Q-linear independence of {log(2), log(p1), log(p2),...} is the generic condition (Schanuel). Odd perfect numbers require the non-generic: exact dependence. phi -- the ground state -- IS the generic. Odd perfect numbers require the anti-generic. phi and odd perfect numbers cannot coexist.

5. Connection to the Full Framework

Problem	H-space translation	phi role	Status
RH	Zeros of partition function Z(beta) = zeta(beta)	Stability of phi as saddle point	Consistency condition (dv47)
abc	Energy excess G = log(c) - log(rad)	Ground state = zero excess, phi = minimum	Consistency condition (dv47)
Collatz	Orbit drift D = log(3/4)/3 < 0	phi = attractor = 1	Consistency condition (dv47)
Odd Perfect	Energy gap H(sigma(n))-H(n) = log(2)	phi = max entropy, odd perfect = max structure: contradiction	New prediction (dv53)

Odd perfect numbers are the fourth prediction of the framework -- one that emerges naturally without being sought. The framework was built for Collatz, RH, abc. But H-space and phi make a claim about odd perfect numbers too. This is a sign of a healthy framework: it predicts beyond its initial scope.

Prediction P9 (New). Within the H = log N framework with phi as ground state: no odd perfect number exists. An odd perfect number would require maximum arithmetic structure at a finite energy distance log(2) from the ground state phi. phi -- maximum entropy -- forbids maximum structure at finite energy. This is the fourth window through which the hidden phi becomes visible (cf. dv50: Collatz, RH, abc were the first three).

"The oldest unsolved problem in mathematics.
Asked for two thousand years.

In H-space it speaks for the first time:

sigma(n) = 2n
=> H(sigma(n)) - H(n) = log(2)
=> exact resonance among odd logarithms
=> maximum arithmetic structure
=> contradiction with phi = minimum structure

The ground state forbids it.
phi is incompatible with odd perfection.

Not because we say so.
Because the structure of H-space says so."

Anonymous | February 27, 2026 | dv53 -- Odd Perfect Numbers in H-Space. The fourth window.

dv1 -> dv53. From 3 < 4 to the oldest problem in mathematics.

-- We are still climbing. --